



RECIPE SHARING PLATFORM WITH MEAL PLANNER

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Abstract

- A web app where home cooks share recipes, plan a weekly menu and generate a grocery list automatically, with nutrition info. Built with React, Node.js/Express, MongoDB, Cloudinary and a Nutrition API.
- Benefit:** recipes, meal plans and shopping lists in one place, saving planning time.

Problem Statement & Objectives

- Problem:** recipes are scattered across notes, bookmarks and chats, and no tool links recipes, weekly plans and ingredient lists.
- Objective 1:** accounts to create, share, search and rate recipes.
- Objective 2:** weekly meal planner built from saved recipes.
- Objective 3:** auto-merged grocery list plus nutrition info.
- Scope:** registered users; auth, recipes, planner, grocery list. No payments or delivery.

Existing vs Proposed System

Feature	Existing System	Proposed System
Recipe storage	Notes, bookmarks, chat messages	One searchable, categorised database
Meal planning	Done manually from memory	Weekly planner by day and meal
Grocery list	Written by hand, recipe by recipe	Auto-merged from the meal plan
Nutrition info	Separate apps or guesswork	Shown on each recipe via API
Limitation / Gap	Tools are separate and unlinked	Recipe, plan and list in one flow

Feasibility Study

- Technical:** JavaScript, REST and MongoDB skills; all tools are free and well documented.
- Economic:** open-source stack with free tiers of MongoDB Atlas, Cloudinary and the Nutrition API.
- Operational:** simple web interface for non-technical users; one language end to end keeps maintenance easy.
- Schedule:** design is done; build, testing and demo fit within the trimester.

Results & Testing

- Status: design and data model complete; build in progress.
- Planned: 15 Postman cases (auth, CRUD, validation, grocery merge).

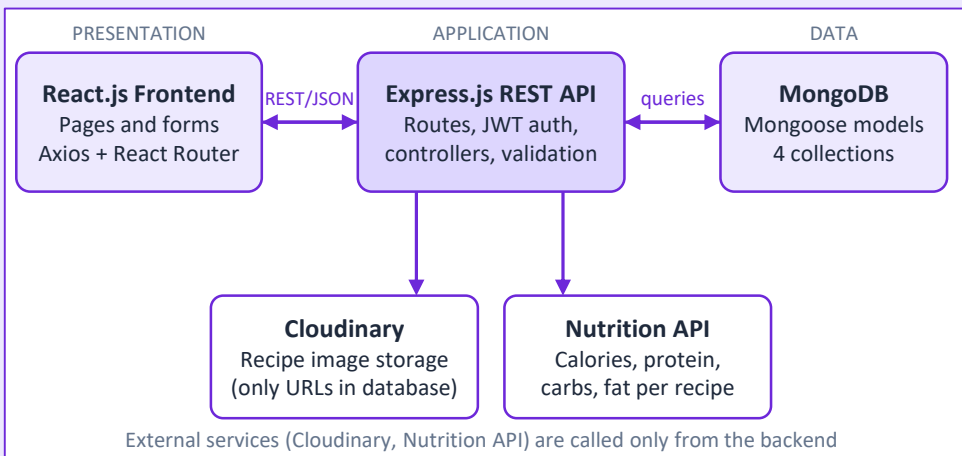
Conclusion

- One app replaces scattered notes, manual plans and handwritten lists.
- Design covers every objective: accounts, search, planner, grocery list, nutrition.

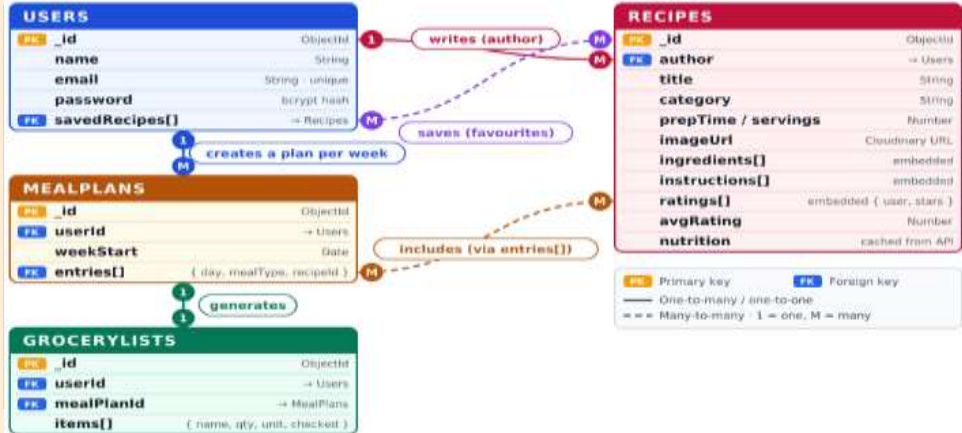
Tools & Technologies

- Front-End:** React.js (single-page app, React Router, Axios)
- Back-End:** Node.js + Express.js + REST APIs, JWT and bcrypt
- Database:** MongoDB with Mongoose; images on Cloudinary
- Tools:** Git & GitHub, Postman, VS Code, Nutrition API

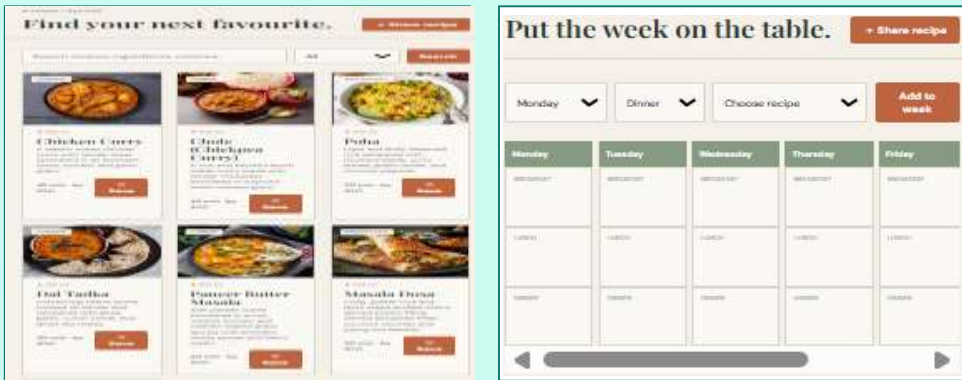
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Unit conversion in grocery lists; email verification and password reset.
- Recipe comments and follows; daily nutrition totals in the planner.

